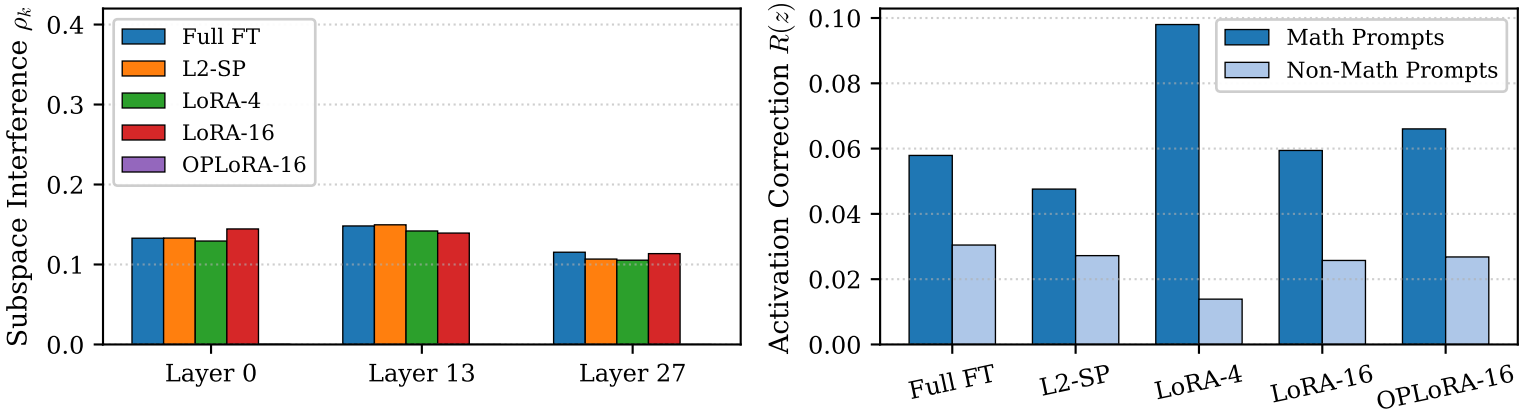
(a) Output Subspace Overlap $\rho_k$ on $W_{\text{down}}$
Subspace Interference $\rho_k$
0.0
0.1
0.2
0.3
0.4
Full FT
L2-SP
LoRA-4
LoRA-16
OPLoRA-16
Layer 0
Layer 13
Layer 27
(b) Layer 13 Activation Impact $R(z)$
Activation Correction $R(z)$
0.00
0.02
0.04
0.06
0.08
0.10
Math Prompts
Non-Math Prompts
Full FT
L2-SP
LoRA-4
LoRA-16
OPLoRA-16